

Mutual Fund Analytics & Performance Evaluation

Project Overview

This project is a data-driven Mutual Fund Analytics system designed to analyze mutual fund performance using real-world financial datasets. It focuses on evaluating risk, returns, and investment behavior using structured data pipelines and visualization tools.

The project includes end-to-end workflow from data ingestion to dashboard creation.

Objectives

- Analyze mutual fund performance using historical data
 - Evaluate risk and return using financial metrics
 - Build an ETL pipeline for structured data processing
 - Generate insights using EDA and visualization
 - Create an interactive Power BI dashboard
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Project Structure

- data/ → Raw and processed datasets
 - notebooks/ → Jupyter notebooks for analysis
 - data_ingestion.py → Data collection script
 - data_cleaning.py → Data preprocessing script
 - live_nav_fetch.py → Live NAV data fetcher
 - bluestock_mf.db → SQLite database
 - reports/ → Documentation files
 - sql/ → SQL queries
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Tech Stack

- Python (Pandas, NumPy)
 - SQLite
 - Jupyter Notebook
 - Power BI
 - SQL
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ETL Pipeline

Raw Data → Extract → Clean → Transform → Analyze → Dashboard

- Data extracted from multiple CSV files
 - Cleaned using Python scripts
 - Feature engineering performed for financial metrics
 - Data loaded into CSV & SQLite database
 - Visualized using Power BI dashboard
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Key Features

- CAGR, Sharpe Ratio, Alpha, Beta, VaR, CVaR analysis
 - Portfolio performance evaluation
 - Risk vs return comparison
 - Category-wise mutual fund analysis
 - Investor behavior insights
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Dashboard

An interactive Power BI dashboard was created to visualize:

- Fund performance
 - Risk analysis
 - Category distribution
 - Investment trends
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How to Run

1. Clone the repository
 2. Install dependencies using `requirements.txt`
 3. Run `data_ingestion.py`
 4. Run `data_cleaning.py`
 5. Open Jupyter notebooks for analysis
 6. View Power BI dashboard file
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Author

Kashish Khan

Bluestock Internship Capstone Project

Thank You

Thank you for reviewing this project.